Sicheng Wang

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Education

New York University, Courant Institute of Mathematical Sciences New York, NY
B.S. in Honors Mathematics; Minor in Computer Science Expected Dec. 2026
GPA: 3.87/4.00

Research Interests

Random walks and electrical networks; discrete potential theory; Gaussian free field; branching random walk; cover times; extrema of log-correlated fields; probabilistic combinatorics.

Independent Study and Mentored Reading

Random Walks and Electrical Networks 2025–2026
Mentor: Prof. Ofer Zeitouni, NYU Courant

- Completed a guided reading of Doyle and Snell's *Random Walks and Electric Networks*, focusing on the connection between reversible Markov chains, harmonic functions, electrical networks, effective resistance, and discrete potential theory.
- Studied core tools including shorting and cutting principles, Rayleigh monotonicity, Thomson's principle, network flows, and recurrence/transience criteria for random walks.
- Worked on an explicit problem for simple random walk on $\mathbb{Z}^2$: computing the probability that a walk started at the origin hits $(0, 1)$ before returning to the origin; obtained the value $1/2$ through a rigorous argument.
- Later revisited the same problem using Green-function and potential-function methods, connecting the electrical-network viewpoint to analytic techniques related to the discrete Gaussian free field.

Gaussian Free Field, Branching Random Walk, and Cover Times 2026–Present
Mentor: Prof. Paul Bourgade, NYU Courant

- Currently studying the discrete Gaussian free field, branching random walk, and related extreme-value phenomena through mentored reading.
- Reading Biskup's PIMS lecture notes on the Gaussian free field and Zeitouni's notes on Gaussian processes and related probabilistic methods.
- Studied finite-volume DGFF constructions via the discrete Dirichlet energy, Green-function covariance, and Hilbert-space / inverse-Laplacian perspectives.
- Current goal: build toward Ding–Lee–Peres' work relating graph cover times, blanket times, Gaussian free fields, and Talagrand's majorizing measures theory.

Talks and Presentations

"Connections Between Random Walks, Electrical Networks, and the Discrete Gaussian Free Field"
Hudson River Undergraduate Mathematics Conference Apr. 25, 2026

- Presented the connection between simple random walks, harmonic functions, electrical networks, effective resistance, and the discrete Gaussian free field.
- Emphasized how electrical-network tools such as shorting/cutting, Rayleigh monotonicity, and Thomson's principle can be used to analyze recurrence, transience, and hitting probabilities.

"Techniques to Analyze Random Walks on the Lattice"
Probabilistic Combinatorics Reading Seminar, Courant Institute, NYU Spring 2026

- Gave a two-hour presentation in a small Ph.D.-level reading seminar on probabilistic combinatorics.
- Presented electrical-network and combinatorial methods for analyzing random walks on lattices, including harmonic functions, voltage/current interpretation, effective resistance, Thomson's principle, Rayleigh monotonicity, and shorting/cutting arguments.
- Participated throughout the semester in talks on Ding–Lee–Peres' cover-time theorem, extremal graph theory, random graphs, discrete martingales, the second moment method, the Lovasz Local Lemma, correlation inequalities, Ramsey-type theorems, statistical physics methods in combinatorics, and Szemeredi's regularity lemma.

Research Notes and Ongoing Projects

Random Walks, Electrical Networks, and Potential Theory

- Writing notes on the connection between simple random walks, electrical networks, harmonic functions, effective resistance, and discrete potential theory.
- Includes a worked analysis of the $\mathbb{Z}^2$ hitting problem

$$\mathbb{P}_0(\tau_{(0,1)} < \tau_0^+) = \frac{1}{2}$$

  using electrical-network and potential-function methods.

Discrete Gaussian Free Field and Cover Times

- Ongoing study project on the discrete Gaussian free field, branching random walk, and cover-time theory.
- Current focus: understanding the role of Green functions, log-correlated fields, and majorizing measures in the Ding–Lee–Peres framework.

AI-Assisted Mathematical Research Workflow

- Developed a personal workflow for distilling mathematical reading notes into reusable proof templates, conceptual summaries, and LLM-agent "skills."
- Applied this system to ongoing study in random walks, electrical networks, Gaussian free field, branching random walk, and cover-time theory.
- Focused on preserving mathematical rigor by using agents for decomposition, analogy search, concept retrieval, and exposition, while keeping final proof verification under human control.

Selected Coursework

Honors Analysis I & II; Honors Algebra II; Algebra; Honors Probability; Applied Complex Variables; Applied Partial Differential Equations; Numerical Analysis; graduate coursework: Alternative Data, Cryptocurrencies & Blockchains (Courant, MATH-GA). Computer Science minor: Data Structures & Algorithms, Design & Analysis of Algorithms.

Technical and Research Skills

Mathematical Writing: LaTeX, Beamer, research notes, seminar slides.
Programming and Computing: Python, C++, SQL, Linux, Git, pandas, NumPy.
AI-Assisted Research: LLM-assisted literature review, proof decomposition, note distillation, mathematical exposition, and agent-based research workflow design.

Quantitative Modeling Experience

Co-Founder / Quantitative Research & Risk 2021–Present
Private Proprietary Crypto Trading Partnership

- Co-managed a proprietary market-neutral crypto trading operation, focusing on probabilistic modeling, risk control, derivatives hedging, and event-driven arbitrage.
- Developed practical experience with stochastic modeling, tail-risk management, market microstructure, and prediction-market order books.
- Current applied research interests include passive market making, adverse selection, and information aggregation in binary prediction markets.

Honors & Distinctions

- 2022 USAMO Qualifier.